

Appendix E - Music Note and Frequency Tables

Each audio engine programs pitch through either a direct Hertz value, a period/divisor, or a chip-specific frequency register. This appendix gives the formula for each engine and a 12-note octave-4 table.

The reference frequencies are:

- **SoundChip / SFX / WAV / MOD / Amiga Paula DMA:** programmed in Hertz directly (the mixer runs at the system output sample rate). No divisor formula applies.
- **PSG / AY-3-8910:** master clock 2,000,000 Hz, period $\text{clock} / (16 * f)$.
- **SN76489:** master clock 3,579,545 Hz (NTSC), period $\text{clock} / (32 * f)$.
- **SID:** master clock 985,248 Hz (PAL), period $f * 16777216 / \text{clock}$.
- **TED audio:** PAL master clock 886,724 Hz, sound clock 110,840 Hz, register $1024 - \text{sound_clock} / f$.
- **POKEY:** master clock 1,789,773 Hz (NTSC) divided by 28, divisor $\text{clock} / 28 / f - 1$.
- **MIDI/MUS:** use MIDI note numbers. The player converts note 69 to A4 = 440 Hz and applies pitch bend internally.

E.1 Octave 4 (middle C through B), equal temperament A4 = 440 Hz

Note	f (Hz)	PSG period	SN76489 period	SID period	TED register	POKEY divisor
C	261.63	478	427	4455	600	243
C#	277.18	451	403	4720	624	229
D	293.66	426	381	5001	647	216
D#	311.13	402	359	5298	668	204
E	329.63	379	339	5612	688	192
F	349.23	358	320	5947	707	182
F#	369.99	338	302	6300	724	171
G	392.00	319	285	6675	741	162
G#	415.30	301	269	7073	757	153
A	440.00	284	254	7493	772	144
A#	466.16	268	240	7939	786	136
B	493.88	253	226	8410	800	128

E.2 Extending the table

To move up one octave (double the frequency): halve the PSG and SN76489 periods, double the SID period, halve the POKEY divisor, and for TED halve the internal divisor:

```
newTED = 1024 - INT((1024 - oldTED) / 2)
```

To move down one octave (halve the frequency): double the PSG and SN76489 periods, halve the SID period, double the POKEY divisor, and for TED double the internal divisor:

$$\text{newTED} = 1024 - 2 * (1024 - \text{oldTED})$$

The relation is exact to within rounding error. Octave shifts of more than four become noticeably flat or sharp on the small-divisor chips (SN76489, POKEY, and high TED values) and need a tempered correction table for accurate music.

E.3 SoundChip, MIDI/MUS, and modern engines

The SoundChip channels and the sample-based engines (WAV, MOD, SFX, Amiga Paula DMA) take frequency in Hertz directly through the FREQ register or period model owned by that engine. A program writes `int ( round ( f ) )` into a SoundChip FREQ register and the channel plays at that frequency. No table is needed: middle C is 262, A4 is 440.

MIDI/MUS does not expose note pitch as a divider register. The file stores note numbers, programme changes, volume/expression controllers, tempo, and pitch bend. The RawlandMini synth converts those note numbers to frequency internally, with note 60 as middle C and note 69 as A4.

E.4 Tuning notes

The numbers in section E.1 are calculated, not measured. The real-silicon SID, PSG, and POKEY chips drift with temperature and with the precise master-clock crystal; an Intuition Engine program that needs the same pitch on every machine should derive its own period from the engine clock at startup rather than hard-coding the values above.

The reference table assumes equal temperament. A program that wants just intonation, meantone, or other historical tunings generates the period table from the desired ratios using the same divisor formula. The engines have no awareness of a "scale": each channel plays whatever pitch its period register encodes.